

PROGRAMME : Diploma Programme in CE / ME / PO / EE / IF / CM / EL / AE / DD / ID / MK
COURSE : Environmental Education and Sustainability **COURSECODE:231302**

LEARNING AND ASSESSMENT SCHEME:

Learning Scheme					Credits	Assessment Scheme										Total Marks	
Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory			Based on LL and TL				Based on self Learning			
										Practical							
CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
03	00	00	01	04	02	02	Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	125
							30	70* #	100	40	--	--	--	--	25	10	

IKS Content:02 Hrs, Total Learning Hours for Term:03 Hrs, # Indicates Online Examination, @ Internal assessment.

1.0 RATIONALE:

The world today is facing the biggest challenge of survival. Degradation of ecosystem, depletion of natural resources, increasing levels of pollution pose major threat to the survival of mankind. The need of the hour, therefore, is to concentrate on the area of environmental aspects, which shall provide an insight into various environment related issues. Environmental studies are an interdisciplinary academic field that integrates physical, chemical and biological sciences, with the study of the environment. It provides an integrated, quantitative, and interdisciplinary approach to the study of environmental system & gives an insight into solutions of environmental problems.

2.0 INDUSTRY / EMPLOYER EXPECTED OUTCOME:

Empower the students to apply the concept of sustainability within the workplace, fostering environmentally responsible practices that contribute to long-term ecological, social, and economic well-being. This empowerment extends beyond isolated environmental considerations to encompass social responsibility and economic viability, aligning with a holistic vision of sustainable business practices for long-term prosperity.

3.0 COURSE OUTCOMES:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

- 1) Develop Public awareness about environment.
- 2) Select alternative energy resources for Engineering Practice.
- 3) Conserve Ecosystem and Biodiversity.
- 4) Apply techniques to reduce Environmental Pollution.
- 5) Manage social issues and Environmental Ethics as lifelong learning.

4.0 COURSE DETAILS:

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-I Environment	1a. Discuss the scope of environment. 1b. Describe the various types of environment. 1c. Discuss about the need of public awareness about environment. 1 d. Describe environmental	1.1 Definitions, need of environmental studies. 1.2 Segments of environment- Atmosphere, Hydrosphere Lithosphere, Biosphere. 1.3 Environmental Issues - Green house effects, Climate change,	06

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
	issues. 1 e. Describe the importance of 'environment studies.	Global warming, Acid rain Ozone layer depletion, Nuclear accidents. 1.4 Concept of 4R (Reduce, Reuse, Recycle and Recover), 1.5 Public awareness about environment	
Unit- II Energy Resources	2a. List various natural resources. 2b. Describe Renewable Non-renewable and Cyclic resources. 2c. State the causes and effects of depletion of resources. 2d. State advantages and disadvantages of forms of energy. 2e. Select appropriate solutions of efficient use of energy. 2f. State the impacts of overuse of natural resources.	2.1 Natural Resources - Forest Resources, Water Resources, Energy Resources, Land resources, Mineral resources. 2.2 Renewable, Non-renewable and Cyclic Resources.(IKS-use of renewable energy sources in ancient India) 2.3 Causes and effects of depletion of resources. 2.4 Energy forms (Conventional and non- conventional). 2.5 Present global energy use and future demands. 2.6 Energy conservation. 2.7 Over use of natural resources and its impacts on environment.	10
Unit- III Ecosystem and Biodiversity	3a. State the aspects and division of ecosystem. 3b. State the general Characteristics and function of ecosystem. 3c. List levels of biodiversity. 3d. Enlist the endangered species. 3e. Describe value of biodiversity. 3f. Suggest methods for biodiversity conservation	3.1 Ecosystem - Definition, Aspects of ecosystem, Division of ecosystem, General characteristics of ecosystem, Functions of ecosystem. 3.2 Biodiversity - Definitions, Levels, Value and loss of biodiversity. 3.3 Biodiversity assessment initiatives in India. 3.4 Threats and Hotspots of biodiversity. 3.5 Conservations of biodiversity objects, various laws.	08
Unit- IV Environmental Pollution	4a. Define pollution. 4b. State the sources of pollution. 4c. State the effects of land pollution on environment and lives. 4d. State various units and their functions of water treatment plant.	4.1 Definition of pollution, types- Natural & Artificial (Man-made). 4.2 Soil / Land Pollution - Causes and effects on environment and lives , preventive measures. 4.3 Water Pollution - Sources of water 4.4 (surface and sub surface),	16

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
	4e. State the needs of water conservation. 4f. State the impacts of sewage. 4g. State various units and their functions of sewage treatment plant. 4h. State sources and effects of air pollution. 4i. Describe various methods to prevent air pollution. 4j. State sources and effects of noise pollution. 4k. Describe preventive measures for noise pollution. 4l. State characteristics of solid waste. 4m. State the impacts of solid waste. 4n. Describe incineration, RDF and sanitary landfilling. 4o. State the standards limiting/controlling values of various types of pollution.	sources of water pollution, on environment and measures. BIS water quality standards, flow diagram of water treatment plant, Water conservation.(IKS- Ancient water conservation practices) 4.5 Wastewater - Generation (domestic and industrial), Impacts, flow diagram of sewage treatment plant, CPCB norms of sewage discharge. 4.6 Air pollution - Causes, effects, prevention, Ambient air quality standards. 4.7 Noise pollution - Sources, effects, prevention, noise levels at various zones of the city. 4.8 Municipal Solid Waste, Bio-medical waste and E-waste – Sources, generation, characteristics, effects, and methods to manage	
Unit-V Social Issues and Environmental Education	5a. Elaborate article (48-A) and (51-A (g)) 5b. 5b. Enlist various acts on environment and its provisions. 5c. 5c. State the roles and responsibilities of CPCB. 5d. 5d. Define sustainable development, and EIA. 5e. 5e. Describe rain water harvesting and groundwater recharge. 5f. 5f. Differentiate between formal and non formal education	5.1 Article (48-A) and (51-A (g)) of Indian Constitution regarding environment, Environmental protection and prevention acts, CPCB and MPCB norms and responsibilities, The role of NGOs. 5.2 Concept of sustainable development, EIA and environmental morality. 5.3 Management Measures - Rain Water harvesting, Ground water recharge, Green Belt Development, Use of Renewable energy, water shed management, interlinking of rivers. 5.4 Role of information technology in environment and human health	08
		TOTAL	48

5.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

Unit No.	Unit Title	Distribution of Theory Marks			
		R Level	U Level	A and aboveLevels	Total Marks
I	Environment	4		6	10
II	Energy Resources	4	8	4	16
III	Ecosystem and Biodiversity	4	4	4	12
IV	Environmental Pollution	8	8	4	20
V	Social Issues and Environmental Education	4	4	4	12
	TOTAL	24	24	22	70

Legends: R = Remembrance (Knowledge); U= Understanding; A= Application and above levels (Revised Bloom's taxonomy)

Note: This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary slightly from above table.

6.0 LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES: Not Applicable

7.0 SELF LEARNING: (Assignment/Activities for Specific Learning/Skill Development/Online Courses/Micro projects

Other than the classroom and laboratory learning, following are the suggested Student-related co-curricular activities. This can be undertaken to accelerate the attainment of the various outcomes in the course. Students should complete minimum 80% of the below mentioned activities.

1. Conduct a waste audit in the college premises.
2. Prepare poster about global warming and ozone depletion.
3. Develop a recycling campaign to raise awareness and write report on it.
4. Encourage students to propose and lead sustainability initiatives in campus.
Examples include community gardens, waste reduction projects, or energy-saving competitions.
5. Report on Use of natural resources and overexploitation of Resources.
6. Write important provisions of Acts related to Environment/ Air (Prevention and Control of Pollution) Act/Water (Prevention and Control of Pollution) Act/ Wildlife Protection Act/ Forest Conservation Act.
7. Collect articles regarding Global Warming, Climate Change.
8. Visit to Water conservation site / Watershed management site / Rainwater harvesting site and writing a report on it.
9. Report on structure and functions of ecosystem.
10. Study organic farming/ Vermiculture /biogas plant and writing a report on it.
11. Video Demonstration to Renewable / Non-renewable (wind mill, hydropower station, thermal power station)/ resources of energy and write report on it.
12. Survey on solid waste management and write report on it.

8.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course:

1. Massive open online courses (*MOOCs*) may be used to teach various topics/sub topics.
2. About *15-20% of the topics/sub-topics* which is relatively simpler or descriptive in nature is to be given to the students for *self-directed learning* and assess the development of the COs through classroom presentations.
3. Arrange various communication activities of real life situation.
4. Use Flash/Animations to explain various topics.
5. Guide micro-projects in group as well as individually.

9.0 ASSESSMENT METHODOLOGY

1) Formative Assessment

FA-TH represents the class test of 30 marks conducted during the term in MCQ form in *online/offline* mode.

2) Summative Assessment

SA-TH represents the summative assessment conducted at the end of the term in MCQ form in *online* mode.

3) Self learning includes micro projects/ assignments /other activities.

10.0 LEARNING RESOURCES:

A) Books:

Sr.No.	Author	Title of Book	Publication
1	Basic Environmental Sciences	Michael Allaby	Routledge Publication, 2nd Edition, 2000, ISBN: 0-415-21176-X
2	Environmental Science	Y. K. Singh	New Age International Publishers, 2006, ISBN: 81-224-2330-2
3	Environmental Studies	Erach Bharucha	University Grants Commission, New Delhi
4	Environmental Studies	Rajagopalan	Third Edition, Oxford University Press, USA, ISBN: 9780199459759, 0199459754
5	A text book of Environmental Science	Arvind Kumar	APH Publishing New Delhi
6	A text book of Environmental Studies	Shashi Chawla	Tata Me Graw-Hill New Delhi
1	Basic Environmental Sciences	Michael Allaby	Routledge Publication, 2nd Edition, 2000, ISBN: 0-415-21176-X

B) Software/Learning Websites:

1. www.eco-prayer.org
2. www.teriin.org
3. www.cpcb.nic.in
4. www.whatis.techtargget.com
5. www.sustainabledevelopment.un.org
6. www.conserve-energy-future.com
7. www.indiaenvironmentportal.org.in

C) Major Equipment/ Instrument with Broad Specifications:

1. LCD projector.
2. Smart Board with internet.

11. MAPPING MATRIX OF PO's, CO's and PSO's:

Course Outcomes	Programme Outcomes (PO's)							Programme Specific Outcomes (PSO's)		
	1	2	3	4	5	6	7	1	2	3
CO1	L	L	M	L	H	L	M	--	--	--
CO2	L	--	--	L	H	L	M	L	--	L
CO3	L	M	M	L	H	L	M	--	L	--
CO4	L	M	--	--	H	L	M	--	L	L
CO5	L	M	M	M	H	L	M	--	L	L

H: High Relationship, M: Moderate Relationship, L: Low Relationship.

12. End semester online examination (70 marks)

UNIT No	CO	Marks per Unit	Questions (01 mark)	Actual Distribution of Marks
I	CO1	10	10	10
II	CO2	16	16	16
III	CO3	12	12	12
IV	CO4	20	20	20
V	CO5	12	12	12
Total		70	70	70